2022 Minnesota Academic Standards in Mathematics Computer Science Learning Progressions

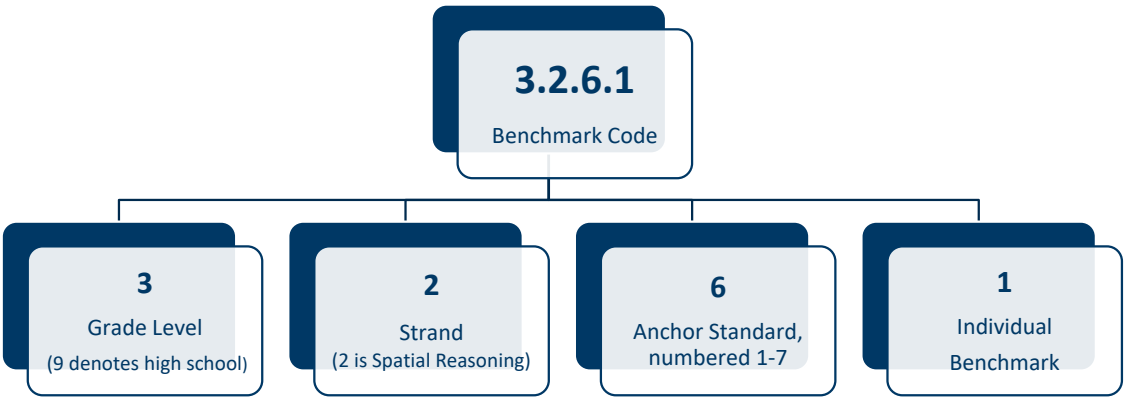
This document show the computer science (cs) integrated benchmarks in the 2022 Minnesota Academic Standards in Mathematics, organized by computer science concept, with their corresponding grade-level benchmarks for grades K–High School. These learning progressions allow teachers, school leaders and administrators, and parents and caregivers to see what students should be learning in their grade level in relation to the grade levels above and below.

The 2022 Minnesota Academic Standards in Mathematics committee looked at connections between the national Computer Science Teachers Association (CSTA) standards and the revised benchmarks to find areas of overlap between the two content areas. The benchmarks that were identified as CS-integrated are marked with a #.

This document only includes the CS-integrated benchmarks and therefore is not a complete listed of the 2022 Minnesota Academic Standards in Mathematics. Importantly, the standard and benchmark information below is the same as can be found in the primary Minnesota Academic Standards in Mathematics documents—available on the MDE Math implementation webpage—but is organized by computer science concept. Other learning progression documents that cover the strands are also available on the MDE Math implementation webpage.

Reading the Mathematics Standards and Benchmarks

Each benchmark has a four-digit code, which includes the grade level, strand, standard and benchmark. Each benchmark is followed by two dimensions; Mathematical Practices (MP) and Benchmark Contexts.

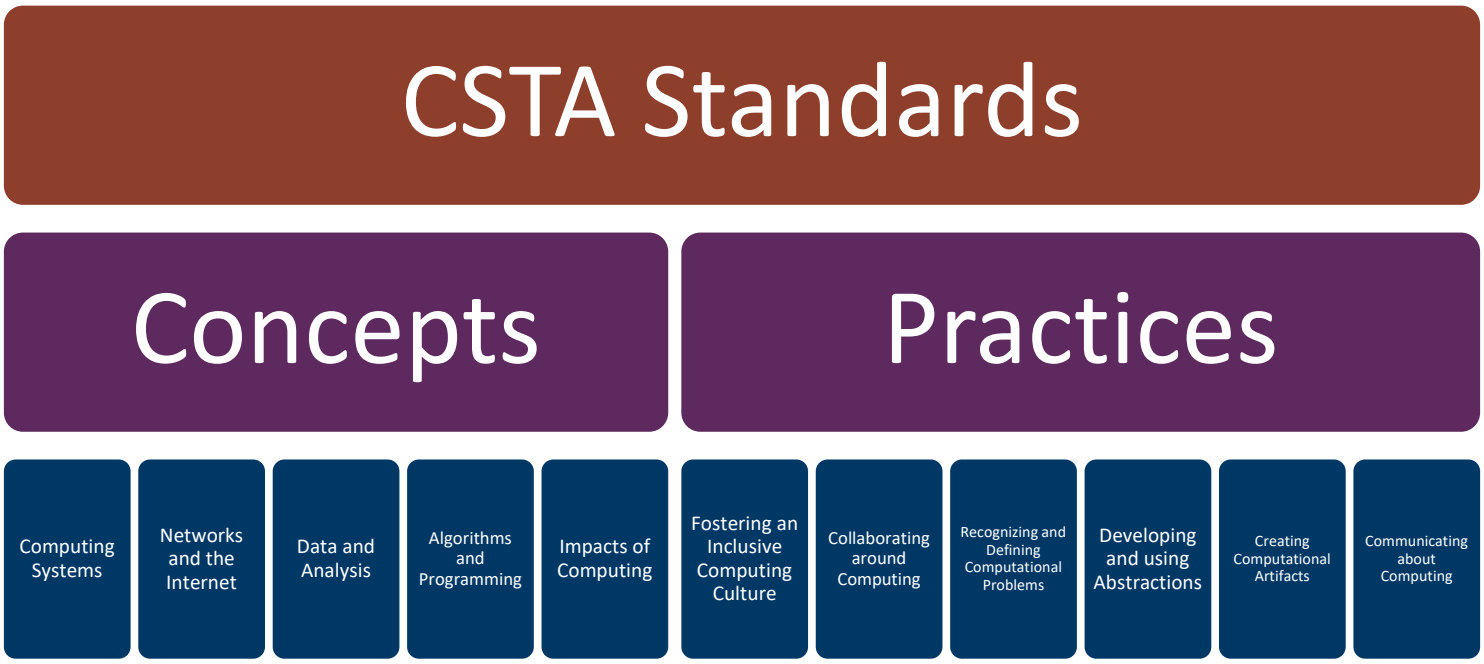


Dimension 1 Mathematical Practices (MP)
• MP1: Make sense of problems and persevere in solving them. • MP2: Reason abstractly and quantitatively. • MP3: Construct viable arguments and critique the reasoning of others. • MP4: Model with mathematics. • MP5: Use appropriate tools strategically. • MP6: Attend to precision. • MP7: Look for and make use of structure. • MP8: Look for and express regularity in repeated reasoning.

Dimension 2 Benchmark Contexts
• ✣ Minnesota Tribal Nations Contexts • $ Financial Literacy Contexts • # Computer Science Contexts • µ Modeling Contexts • ☼ Real-world Contexts

Computer Science Concepts

The 2022 Minnesota Academic Standards in Mathematics committee used the national CSTA standards that were published in 2017. These standards are currently undergoing revision that will be released in Summer 2026. Below is the organization of the concepts and practices included in the 2017 national standards. The computer science -integrated benchmarks in the math standards only connect to the Data and Analysis and Algorithms and Programming Concepts and we encourage teachers to look for connections to the practices.



Data and Analysis

Computing systems exist to process data. The amount of digital data generated in the world is rapidly expanding, so the need to process data effectively is increasingly important. Data is collected and stored so that it can be analyzed to better understand the world and make more accurate predictions.

A. Inference and Models Data can be used to make inferences or predictions about the world. Inferences, statements about something that cannot be readily observed, are often based on observed data. Predictions, statements about future events, are based on patterns in data and can be made by looking at data visualizations, such as charts and graphs. The accuracy of inferences and predictions is related to how realistically data is represented. Many factors influence the accuracy of inferences and predictions, such as the amount and relevance of data collected.

Grade	Code	Benchmark
K▼	0.1.1.1	Notice and describe patterns in data-rich situations. (MP1, MP7) # ☼ ✣
1		No benchmark.
2	2.1.1.4	Make predictions using patterns from data visualizations. (MP7, MP8) # µ
3	3.1.1.5	Critically analyze data visualizations, including frequency tables, bar graphs, picture graphs or number line plots having a variety of scales, to support a claim and solve situations. (MP1, MP3) $ #
4	4.1.1.4	Critically analyze data visualizations, including tables, double bar graphs, timelines, line plots or spreadsheets to support a claim and solve contextual situations. (MP3, MP7) $ # µ
5	5.1.1.4	Critically analyze data visualizations using measures of center and variability, including but not limited to double-bar graphs, line graphs and line plots to support a claim and solve situations. (MP3, MP7) $ #
	5.1.2.2	Use a frequency table to record results from an experiment to make predictions. Place predictions on a number line from 0 to 1. (MP4) # ☼
6	6.1.2.3	Calculate experimental probabilities from experiments where the theoretical probability is known, recognizing that there may be differences between theoretical and experimental probability. Represent the probabilities as percentages, fractions and decimals between 0 and 1 inclusive. Use experimental probabilities to make predictions when actual probabilities are unknown. (MP4) # ☼
7	7.1.2.2	Approximate the probability of a chance event, where the theoretical probability is unknown, by collecting data and observing its long-run frequency. Represent the probabilities as percentages, fractions and decimals between 0 and 1 inclusive. Use approximate probabilities to make predictions when actual probabilities are unknown. (MP4) # ☼
8	8.1.1.4	Use the equation of a linear model to solve situations in the context of bivariate measurement data. Interpret the slope and intercepts in context of the variables. (MP4, MP6) $ #
8	8.1.1.6	Compare and communicate competing explanations for data trends observed, considering the reasonableness of the model's predictions and correlation versus causation. (MP3, MP6) $ #

Grade	Code	Benchmark
8	8.3.7.2	Analyze visual patterns to distinguish between linear and non-linear patterns. For linear patterns, describe how a pattern is changing, name the nth term and write an equation to generalize the nth term. (MP1, MP7) ✣ # ☼
8	8.3.7.5	Explain how changes to the values m or b in the linear function f(x) = mx + b affect the graph of the function. Use graphing technology to examine these effects. Recognize that the graph of the linear equation y = mx + b comes from b units translation of y = mx graph. (MP1, MP5) µ # ☼
High School	9.1.1.8	Use data from a random sample to draw inferences about a population with an unknown characteristic of interest. Generate multiple samples or simulated samples of the same size to gauge the variation in estimates or predictions. (MP1, MP4) # ☼
High School	9.1.1.11	Identify, create and compare statistical models with linear and exponential functions, including linear regression. Assess the reasonableness of model fit using residuals and correlation coefficients. (MP4) # µ
High School	9.1.2.2	Describe events as subsets of a sample space using characteristics of the outcomes. Use Venn diagrams, tables or lists to depict events and compute event sizes. Create representations to find unions, intersections and complements of events. (MP1) #
High School	9.1.2.3	Apply probability concepts, such as intersections, unions and complements of events, and conditional probability and independence, to calculate probabilities and solve situations. (MP4) # ☼
▲ High School	9.3.7.1	Represent and solve situations in various contexts, including financial literacy, using systems of linear equations, systems of linear inequalities and exponential and quadratic functions. (MP4) $ # ☼

B. Collection, Visualization, and Transformation Everyday digital devices collect and display data over time. The collection and use of data about individuals and the world around them is a routine part of life and influences how people live. Data can be displayed for communication in many ways. People use computers to transform data into new forms, such as graphs and charts. People select aspects and subsets of data to be transformed, organized, clustered, and categorized to provide different views and communicate insights gained from the data.

Grade	Code	Benchmark
K▼	0.1.1.2	Organize objects, draw pictures, or use tally marks to represent data and communicate observations. (MP3, MP6) # µ
1	1.1.1.2	Collect and use data to consider and decide what data will answer a question. Represent the data as a drawing, tally marks, frequency bar graph and digitally communicate observations. (MP1, MP3) # µ
2	2.1.1.3	Collect and use data to consider and decide what data will answer a question. Represent the data as drawings, picture graphs, dot plots (a.k.a. line graphs or line plots) and with technology. Communicate observations. (MP3, MP5) # µ

Grade	Code	Benchmark
3	3.1.1.3	Collect and organize data to answer a statistical question using various tools and addressing missing or incomplete data. Represent data in a variety of ways including technology. (MP1, MP5) # µ ☼
4	4.1.1.2	Collect and organize data to answer a statistical question, analyze variability and address missing, incomplete and bias in data. Represent data in a variety of ways, including technology. (MP1, MP5) #
5	5.1.1.3	Collect and organize data to answer statistical questions and analyze measures of center (mean and median) and variability (range). Represent data in a variety of ways, including technology. (MP5) # µ
5	5.1.1.5	Compare and contrast different data displays to determine how the visualizations impact analysis and interpretation. (MP3, MP8) #
6	6.1.1.2	Design and conduct investigations and experiments to gather data, while considering cultural perspectives, to answer statistically investigative questions considering variability and justifying choice of variables. (MP3, MP6) ✜ $ # ☼
6	6.1.1.4	Create a visualization about a data set to describe patterns, highlight relationships or illustrate features of the distribution of the data to answer or help answer their statistically investigative question. Visualizations should represent the data in appropriate ways, including tables, dot plots, stem-and-leaf plots, histograms and box plots while incorporating any other relevant information that helps to tell a story about the data. (MP5, MP6) #
7	7.1.1.5	Create a visualization about a data set, organizing and presenting the data in appropriate ways, including in tables, circle graphs and histograms, and incorporating any other relevant information that helps to tell a story about the data. (MP5, MP6) # µ
7	7.1.2.6	Find probabilities of compound events using organized lists, tables, tree diagrams and/or simulation via a computational tool. (MP5) # µ ☼
8	8.1.1.5	Create data visualizations about a data set. Organize and present the data in appropriate ways, including in tables and scatter plots, and incorporate other relevant information that helps to tell a story and support a claim about the data. (MP1, MP6) ✜ $ # µ
8	8.3.6.9	Represent relationships in various contextual situations using systems of linear equations. Solve systems of linear equations in two variables, symbolically and graphically, understanding that the solution corresponds to the point of intersection of their graphs. (MP4, MP5) $ # ☼
▲ High School	9.1.1.15	Identify and explain misleading uses of data along with how to use spreadsheets, tables or graphing technology to recognize and analyze distortions in data displays. Use interactive data visualizations to support and influence different points of view. (MP3) $ #

Algorithms and Programming

An algorithm is a sequence of steps designed to accomplish a specific task. Algorithms are translated into programs, or code, to provide instructions for computing devices. Algorithms and programming control all computing systems, empowering people to communicate with the world in new ways and solve compelling problems. The development process to create meaningful and efficient programs involves choosing which information to use and how to process and store it, breaking apart large problems into smaller ones, recombining existing solutions, and analyzing different solutions.

A. Modularity Complex tasks can be broken down into simpler instructions, some of which can be broken down even further. Likewise, instructions can be combined to accomplish complex tasks.

Grade	Code	Benchmark
K▼	0.3.7.1	Recognize, create, complete and extend simple patterns using shape, color, size, number, sounds and movements. Patterns may be repeating, growing or shrinking. (MP1, MP7) ✣ # ☼
1	1.3.7.1	Create simple patterns using objects, pictures, numbers and rules. Identify possible rules to complete or extend patterns. Patterns may be repeating, growing or shrinking. Calculators can be used to create and explore patterns. (MP5, MP7) ✣ # ☼
1	1.3.5.9	Add and subtract within 20, demonstrating fluency for addition and subtraction within 10. Use strategies such as counting on, making ten, decomposing a number leading to a ten using the relationship between addition and subtraction and creating equivalent but easier or known sums. (MP2, MP4) $ # µ
2	2.3.7.1	Identify, create and describe simple number patterns involving repeated addition or subtraction, skip counting and arrays of objects such as counters or tiles. Use patterns to solve situations in various contexts. (MP7, MP8) ✣ $ # ☼
3		No benchmark at this level.
4		No benchmark at this level.
5	5.2.3.1	Develop, justify and use formulas to determine the area of parallelograms and triangles. Find the areas of polygons that can be decomposed into parallelograms and triangles. (MP3) # µ
6	6.2.3.1	Calculate the measurements of the surface area of rectangular and triangular prisms using appropriate units. Justify the formulas used. Justification may involve decomposition, nets or other models. (MP1, MP3) ✣ # ☼
	6.2.3.2	Calculate the measurement of the volume of prisms (including triangular and nonrectangular prisms) using appropriate units. Justify the formulas used. Justification may involve decomposition or other models. (MP1, MP7) ✣ # ☼
	6.2.4.2	Decompose polygons into triangles to investigate the sum of the interior angles of polygons. (MP1, MP7) #
7	7.1.2.4	Represent sample spaces for compound events by decomposing the events using methods such as organized lists, tables and/or tree diagrams. For an event described in everyday language, identify the outcomes in the sample space that compose the event. (MP5) #
8		No benchmark at this level.
▲ High School	9.2.3.4	Develop, justify and use decomposition to determine the formula for surface area and volume of various three-dimensional figures. (MP7, MP8) ✣ # ☼

B. Variables Programmers create variables to store data values of selected types. A meaningful identifier is assigned to each variable to access and perform operations on the value by name. Variables enable the flexibility to represent different situations, process different sets of data, and produce varying outputs.

Grade	Code	Benchmark
K▼		No benchmark at this level.
1		No benchmark at this level.
2		No benchmark at this level.
3		No benchmark at this level.
4		No benchmark at this level.
5	5.3.6.1	Use relational thinking to find a missing value in an open number sentence with addition and subtraction of fractions and decimal expressions. Determine if the equation is true or false and justify the reasoning. (MP3) $ #
6		No benchmark at this level.
7	7.3.6.3	Evaluate algebraic expressions, including expressions containing rational numbers, absolute value and whole number exponents, by applying computational hierarchy of operations at specified values of their variables. (MP7) #
8		No benchmark at this level.
▲ High School	9.3.5.4	Use matrices to represent and manipulate data and interpret the results in context. Add, subtract and multiply matrices of appropriate dimensions. Multiply matrices by scalars. (MP4, MP5) #

C. Control Computers follow precise sequences of instructions that automate tasks. Program execution can also be nonsequential by repeating patterns of instructions and using events to initiate instructions. Programmers select and combine control structures, such as loops, event handlers, and conditionals, to create more complex program behavior.

Grade	Code	Benchmark
K▼		No benchmark at this level.
1		No benchmark at this level.
2		No benchmark at this level.
3		No benchmark at this level.
4		No benchmark at this level.
5		No benchmark at this level.
6		No benchmark at this level.
7	7.1.2.5	Design and use a simulation within a computational tool to generate frequencies for compound events. #
8		No benchmark at this level.
High School	9.2.4.5	Accurately interpret and use words and phrases such as “if ... then,” “if and only if,” “all” and “not.” Recognize the logical relationships between an “if ... then” statement and its inverse, converse and contrapositive. (MP6, MP7) ✣ # ☼

Grade	Code	Benchmark
High School	9.2.4.6	Analyze the validity of a logical argument and give counterexamples to disprove a statement. (MP1, MP3) ✣ # ☼
High School	9.2.4.7	Construct logical arguments using axioms, definitions, theorems and postulates that clearly justify the reasoning. (MP3, MP6) #
High School	9.2.4.14	Describe the steps used to show various sequences of transformations, applying reflections, rotations, translations and/or dilations of geometric figures that map one figure onto the other. (MP8) # ✣
High School	9.3.7.4	Express the terms in an arithmetic or geometric sequence recursively and by giving an explicit formula. (MP8) #
▲ **High School**	9.3.7.8	Define the compounding of interest n times per year according to a recursive formula. Compare the recursive definition of interest to the recursive definition of a geometric sequence t(n) = r(t(n-1)) . Compare the interest formula A = (1 + r/n) nt to the general form of an exponential function y = a(b)x. Explain the purpose of each part of the interest formula. (MP4, MP5) $ # µ